Exercise: 7

Stomatal Distribution

Aim: To study the distribution of stomata on the upper and lower surfaces of leaves.

Principle: Stomata are tiny microscopic structures present in leaves of all flowering plants. Number and distribution of stomata per unit area in variable in leaves of different plants. A typical stoma consists of a pair of guard cells enclosing an aperture in the centre called the stomatal aperture. Stomata perform two important functions; that of, transpiration and exchange of gases.

Requirements: Leaf samples – *Hibiscus* / Balsam / *Bougainvillea* / *Petunia* / *Cassia* / *Solanum* / any broad leaves of dicot and grass) microscope glass slides, cover slips, water, needle, brush, and petridish / watch glasses.



Peel a dicot leaf and monocot leaf. Dicot leaf to be peeled on both sides to observe upper and lower epidermis. Monocot leaf is to be peeled on any side. Mount the peel on a slide and observe under microscope.

Diagnostic features:

- Stomata are more in number on lower epidermis of dicot leaves and evenly distributed in monocot leaves.
- Stoma is guarded by two bean shaped guard cells.

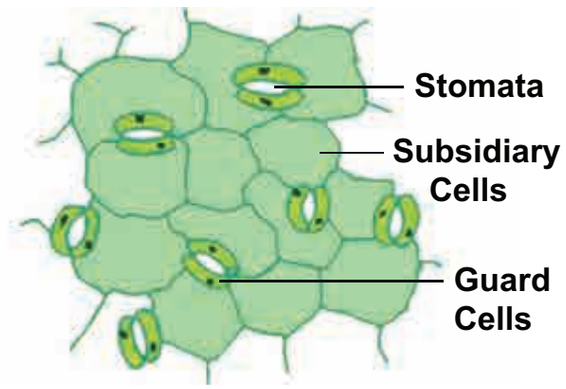


Figure 7: Stomatal distribution

II. Fresh or preserved specimens

Aim: To study and identify the morphology of representative types of Algae, Fungi, Lichen, Bryophytes and Pteridophytes.

Principle: Morphology is the study of the characteristics features of the species. It could be a study of external or internal features. Morphological studies help in identification and classification of organisms.

Requirements: Specimens of *Chara*, Basidiocarp of *Agaricus,* Foliose Lichen, *Funaria, Adiantum*.

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